

Table 5: Quinary Classification Results (F1-Score %)

Model Name	EZS	SCoT	FSR	OFS	FCL	Avg.
<i>Tiny Models</i>						
Gemma 3 27B (ST)	73.02	67.47	72.52	72.81	73.75	71.91
Gemma 3 12B (ST)	72.16	70.17	66.00	64.51	69.36	68.44
Phi-4 (ST)	67.85	66.27	63.00	65.18	67.86	66.03
Mistral-nemo12B (ST)	69.47	67.95	61.94	64.35	67.06	66.15
Mistral-small3.1:24B (ST)	67.73	62.92	67.90	66.49	70.43	67.09
<i>Small Models</i>						
Llama 4 Scout (MoE)	71.98	68.52	68.22	65.15	69.85	68.74
Llama 3.3 (ST)	70.17	69.85	70.80	67.06	64.87	68.55
Mixtral 8x22B (MoE)	71.18	72.30	68.45	71.76	67.85	70.31
Nemotron Super 49B (ST)	67.36	64.58	71.14	64.91	71.22	67.84
Qwen2.5:72B (ST)	72.95	70.44	70.52	69.33	70.05	70.66
<i>Medium Models</i>						
DeepSeek-V2 (MoE)	67.23	66.84	71.19	70.51	61.30	67.41
DeepSeek-Coder-V2 (MoE)	69.65	56.96	70.81	69.27	67.73	66.88
Qwen/Qwen3-235B-A22B (MoE)	70.06	69.43	72.71	67.48	69.32	69.80
Nemotron Ultra 253B (ST)	69.81	69.03	74.00	69.60	70.12	70.51
<i>Large Models</i>						
Llama 4 Maverick (MoE)	72.45	70.31	73.25	67.08	69.09	70.44
<i>Ultra Models</i>						
DeepSeek-R1 (ST)	67.62	65.19	69.16	69.44	68.95	68.07
DeepSeek-V3 (MoE)	68.81	60.59	70.07	68.76	69.87	67.62

Table 6: Quaternary Classification Results (F1-Score %)

Model Name	EZS	SCoT	FSR	OFS	FCL	Avg.
<i>Tiny Models</i>						
Gemma 3 27B (ST)	83.49	81.76	82.61	81.21	86.23	83.06
Gemma 3 12B (ST)	83.80	83.30	81.29	78.48	84.73	82.32
Phi-4 (ST)	80.13	79.14	74.07	80.09	78.99	78.48
Mistral-nemo12B (ST)	73.40	79.96	66.75	76.04	76.17	74.46
Mistral-small3.1:24B (ST)	78.60	81.16	80.03	78.18	77.29	79.05
<i>Small Models</i>						
Llama 4 Scout (MoE)	82.57	81.96	79.54	76.35	82.26	80.54
Llama 3.3 (ST)	85.66	84.33	86.55	80.10	80.41	83.41
Mixtral 8x22B (MoE)	82.48	86.12	79.40	85.05	83.31	83.27
Nemotron Super 49B (ST)	82.98	71.84	86.65	79.20	81.24	80.38
Qwen2.5:72B (ST)	83.18	82.44	78.36	83.45	77.62	81.01
<i>Medium Models</i>						
DeepSeek-V2 (MoE)	79.62	80.14	81.15	82.70	81.04	80.93
DeepSeek-Coder-V2 (MoE)	84.69	77.06	84.51	82.81	84.60	82.73
Qwen/Qwen3-235B-A22B (MoE)	81.79	79.46	82.68	80.36	78.27	80.51
Nemotron Ultra 253B (ST)	81.72	82.35	83.61	81.07	81.47	82.04
<i>Large Models</i>						
Llama 4 Maverick (MoE)	82.48	85.37	83.44	80.78	81.67	82.75
<i>Ultra Models</i>						
DeepSeek-R1 (ST)	75.11	70.02	77.73	78.27	81.68	76.56
DeepSeek-V3 (MoE)	83.61	78.08	83.17	81.22	86.71	82.56

Table 7: Overall Average Performance Across All Tasks (F1-Score %)

Model Name	Quinary Avg.	Quaternary Avg.	Overall Avg.
<i>Tiny Models</i>			
Gemma 3 27B (ST)	71.91	83.06	77.48
Gemma 3 12B (ST)	68.44	82.32	75.38
Phi-4 (ST)	66.03	78.48	72.25
Mistral-nemo12B (ST)	66.15	74.46	70.31
Mistral-small3.1:24B (ST)	67.09	79.05	73.07
<i>Small Models</i>			
Llama 4 Scout (MoE)	68.74	80.54	74.64
Llama 3.3 (ST)	68.55	83.41	75.98
Mixtral 8x22B (MoE)	70.31	83.27	76.79
Nemotron Super 49B (ST)	67.84	80.38	74.11
Qwen2.5:72B (ST)	70.66	81.01	75.84
<i>Medium Models</i>			
DeepSeek-V2 (MoE)	67.41	80.93	74.17
DeepSeek-Coder-V2 (MoE)	66.88	82.73	74.81
Qwen/Qwen3-235B-A22B (MoE)	69.80	80.51	75.16
Nemotron Ultra 253B (ST)	70.51	82.04	76.28
<i>Large Models</i>			
Llama 4 Maverick (MoE)	70.44	82.75	76.59
<i>Ultra Models</i>			
DeepSeek-R1 (ST)	68.07	76.56	72.31
DeepSeek-V3 (MoE)	67.62	82.56	75.09